

Dr. Martin JACQUET

Postdoctoral Researcher

PhD in Robotics
Engineer in Computer Sciences
and Applied Mathematics

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4D allée Verlaine,
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France
28 septembre 1993
Driving license

EXPERIENCES & EDUCATION

Sep 2025 – present



Postdoctoral Researcher, at INRIA - RAINBOW team, Rennes, France

Perception-based physical interaction with aerial robots

With Dr. [Marco Tognon](#) – [ANR-FlyHandyBot](#) project

- Nonlinear control for vision-constrained tasks
- Hybrid/Stochastic NMPC, MPPI
- Integration of onboard lidar-inertial state estimator
- ROS2, GenoM; Python, C++ programming

2020 – present



Temporary Lecturer

- At [ENS Rennes](#) (M1, M2):
 - (2026) 28 hours of lectures and practicals – Nonlinear control
 - (2025) 6 hours of lectures – Aerial robotics: modeling and control
- At [NTNU](#) (M1):
 - (2023) 10 hours of lectures and practicals – ROS development
- At [INP-ENSEEIH](#) (L3, M1):
 - (2021) 31 hours of seminars and practicals – C programming, Android programming
 - (2020) 21 hours of practicals – Android programming

Feb 2023 – Jun 2025



Postdoctoral Researcher, at NTNU - Autonomous Robots Lab, Trondheim, Norway

Perception-based control methods for collision-free navigation in unknown, complex and cluttered environments

With Prof. [Kostas Alexis](#) – [EU-Digiforest](#) project

- Deep learning for latent state/environment representations
- Interplaying neural methods with principled control methods (NMPC, CBF) for provable safety
- Software integration and Autonomy Stack development
- Neural world models, Reinforcement Learning
- ROS2, ROS, PX4; python, C++ programming
- Robotics field-testing campaigns: hardware deployment, mission operator in forest environments

Jan 2019 – Oct 2022



PhD in Robotics, at LAAS-CNRS - RIS group, Toulouse, France

Methods for Online Predictive Control of Multi-rotor Aerial Robots with Perception-driven Tasks subject to Sensing and Actuation Constraints

Advised by Prof. [Antonio Franchi](#) – [ARN-MuRoPhen](#) project

- Perception-constrained NMPC, Multi Aerial Robot Systems
- Active Perception methods, Observation uncertainty estimation
- ROS, GenoM; Python, Matlab and C++ programming
- Involvement in the 2020 [MBZIRC](#) robotics competition: perception-related software programming
→ **Bronze medal** in the Grand Challenge competition

Mar 2017 – Jun 2018



Research Engineer at LAAS-CNRS - RAP group, Toulouse, France

- Collaborative project with the French Wine Institute (Gaillac) and Naïo Technology (Toulouse)
- Deep Learning-based detection of trunks and leaf diseases on wine images and videos
- Semantic segmentation, object detection, spatio-temporal classification
- Python and C++ programming, libraries Keras and TensorFlow

Summer 2016



6 months engineering internship at ATOS, Toulouse, France

Detection and tracking of icebergs in satellite images

- Semantic segmentation, classifications, temporal tracking
- C++ programming, web technologies (API development, SQL database, Docker, Kubernetes)

Summer 2015



3 months research internship at IRIT, Toulouse, France

Search of symmetries for 3D-meshes compression

- Transposition of 2D algorithms in 3D for feature extraction (SIFT)
- Statistical evaluation methodology
- Matlab programming

Summer 2014

Work training period in a Japanese hotel, as a cleaning person

Kamogawa Hotel Mikazuki, Kominato, Chiba, Japan

2013 – 2016

Engineering degree, at INP-ENSEEIH, Toulouse, France

Computer sciences and applied mathematics department



- Audiovisual data processing - Computer vision - 3D Modeling - Virtual Reality
- Mathematics - Statistics/Probabilities - Linear algebra - Optimization

- Computer sciences - Software programming

- School semester at **Polytechnique Montreal**, Canada; Courses in multimedia technologies

- 2 months master's project : Building a 3D model of a scene from a flow of RGBD data

2011 – 2013

Preparatory school, PSI* (Physics and Engineer Sciences)

Lycée Champollion, Grenoble, France

2011

Scientific Baccalauréat

Lycée Pierre Termier, Grenoble, France

SCIENTIFIC PUBLICATIONS

Citation count and data as of 09/07/2026

Total citations: 180

h-index: 7

JOURNAL PUBLICATIONS

IJRR, 2025

–Cited: 1

M. Jacquet, M. Harms, K. Alexis, *Neural NMPC through Signed Distance Field Encoding for Collision Avoidance*

RAL, 2022

–Cited: 22

M. Jacquet, M. Kivits, H. Das, A. Franchi, *Motor-level N-MPC for Cooperative Active Perception with Multiple Heterogeneous UAVs*

RAL, 2021

–Cited: 40

M. Jacquet, A. Franchi, *Motor and Perception Constrained NMPC for Torque-Controlled Generic Aerial Vehicles*

– **Finalist for Best Paper Award** in Unmanned Aerial Vehicles at ICRA'21

RAL is the IEEE reference for timely publications of major results in Robotics

Impact Factor: 5.3 – Journal Citation Indicator: 1.22

IJRR is among the two major venues for comprehensive Robotics publications

Impact Factor: 5.0 – Journal Citation Indicator: 1.36

CONFERENCE PUBLICATIONS

ICRA'25

–Cited: 8

M. Harms, **M. Jacquet**, K. Alexis, *Safe Quadrotor Navigation using Composite Control Barrier Functions*

ICUAS'25

–Cited: 2

N. Misyats, M. Harms, M. Nissov, **M. Jacquet**, K. Alexis, *Embedded Safe Reactive Navigation for Multirotors Systems using Control Barrier Functions*

IROS'24

–Cited: 23

M. Harms, M. Kulkarni, N. Khedekar, **M. Jacquet**, K. Alexis, *Neural Control Barrier Functions for Safe Navigation*

ICRA'24

–Cited: 20

M. Jacquet, K. Alexis, *N-MPC for Deep Neural Network-Based Collision Avoidance exploiting Depth Images*

IROS'22

–Cited: 24

G. Corsini, **M. Jacquet**, H. Das, A. Affi, D. Sidobre, A. Franchi, *Nonlinear Model Predictive Control for Human-Robot Handover with Application to the Aerial Case*

IROS'22

–Cited: 6

M. Jacquet, A. Franchi, *Enforcing Vision-Based Localization using Perception Constrained N-MPC for Multi-Rotor Aerial Vehicles*

AIRPHARO'21

–Cited: 5

G. Corsini, **M. Jacquet**, AE. Jimenez-Cano, A. Affi, D. Sidobre, A. Franchi, *A General Control Architecture for Visual Servoing and Physical Interaction Tasks for Fully-actuated Aerial Vehicles*

ICRA'20

–Cited: 27

M. Jacquet, G. Corsini, D. Bicego, A. Franchi, *Perception-constrained and Motor-level Nonlinear MPC for both Underactuated and Tilted-propeller UAVS*

ICRA and IROS are the two IEEE flagship conferences in Robotics

BOOK CHAPTERS

Springer, 2026 M. Camurri, E. Tomelleri, M. Mattamala, S. Barbas Laina, **M. Jacquet**, *et al.*, *DigiForest: Digital Analytics and Robotics for Sustainable Forestry*, in *Recent Advances in Robotic Perception for Forestry*

EDITORIAL SERVICES

Reviewer Journals: [Nature Communications](#), [T-RO](#), [IJRR](#), [RAL](#)
Conferences: RSS, ICRA, IROS, ICUAS, ECC

Associate Editor IROS (2026)

CO-SUPERVISION OF MASTER'S STUDENTS

Sep 2024 – Dec 2024 Nazar Misyats, ENS Rennes, France
Development of a Safe Embedded Drone Controller based on Control Barrier Functions
Role: Topic definition, theoretical supervision, experimental hardware validation

Jan 2024 – Jul 2024 Ask Espensønn Øren, NTNU, Norway
Multi-modal Learning-based Navigation
Role: Topic definition, technical & theoretical supervision

Jan 2024 – Jul 2024 Mathias Skogli, NTNU, Norway
Reinforcement Learning for Autonomous Safe Navigation in Dynamic and Cluttered Environments Exploiting Dynamics-Aware Embeddings of Depth Images
Role: Topic definition, technical & theoretical supervision

Jan 2020 – Jul 2021 Max Kivits, University of Twente, The Netherlands
Heterogeneous Cooperative Target Tracking using Nonlinear Model Predictive Control
Role: Topic definition, technical supervision, experimental hardware validation

Jan 2020 – Jul 2020 Vamshi Kodipaka, University of Bourgogne, France
Vision based Multi-robot Tracking of Target with Aerial and Ground Robots
Role: Topic definition, technical supervision

MAINTAINED OPEN SOURCE SOFTWARE

Autonomy Stack Comprehensive Unified Autonomy Stack for Robots – Developer – 210 stars

SDF-NMPC Neural MPC for collision avoidance – Main developer, first author of related paper – 47 stars

CCBF Composite CBF for collision avoidance – Main developer, author of related paper – 12 stars

PX4-CBF Embedded impl. of CCBF in [PX4 Autopilot](#) – Developer, author of related paper – 10 stars

All my papers also have an open-source implementation for reproducibility (not actively maintained)

TECHNICAL SKILLS

Scientific Neural Networks, Implicit neural representations
Nonlinear optimal control
Pose estimation, uncertainties, Kalman filtering
Deep learning for image and video processing
Mathematical modeling, physical parameters estimation

Languages Python, C++, C, Matlab

Programming ROS2, ROS, GenoM, PX4, LaTeX, Git, Docker, Docker-compose

Libraries PyTorch, OpenCV, Eigen

Hardware Camera (calibration, data acquisition), Lidars, IMUs, Custom drones & electronics

LANGUAGES

French Native speaker

English C2, fluent written and oral communication; TOEIC obtained in 2015 (940)
Professional daily communication at work (working in international teams since 2019)

Spanish High-school level (B1)

Japanese Advanced Beginner

Norwegian	Beginner
Italian	Beginner

VARIOUS INFO

Hobbies	Hobby software programming, cycling, brewing, tabletop role-playing, video games, cinema, photography, science-fiction
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